

STUDENT LABORATORY MANUAL

Module 2: Introduction to Computer Graphics and Visual Computing

20 Guided Examples • 20 Practice Exercises • 20 Outputs

Course: CS0045 – Computer Graphics and Visual Computing

Companion: Module 2 Student Guide

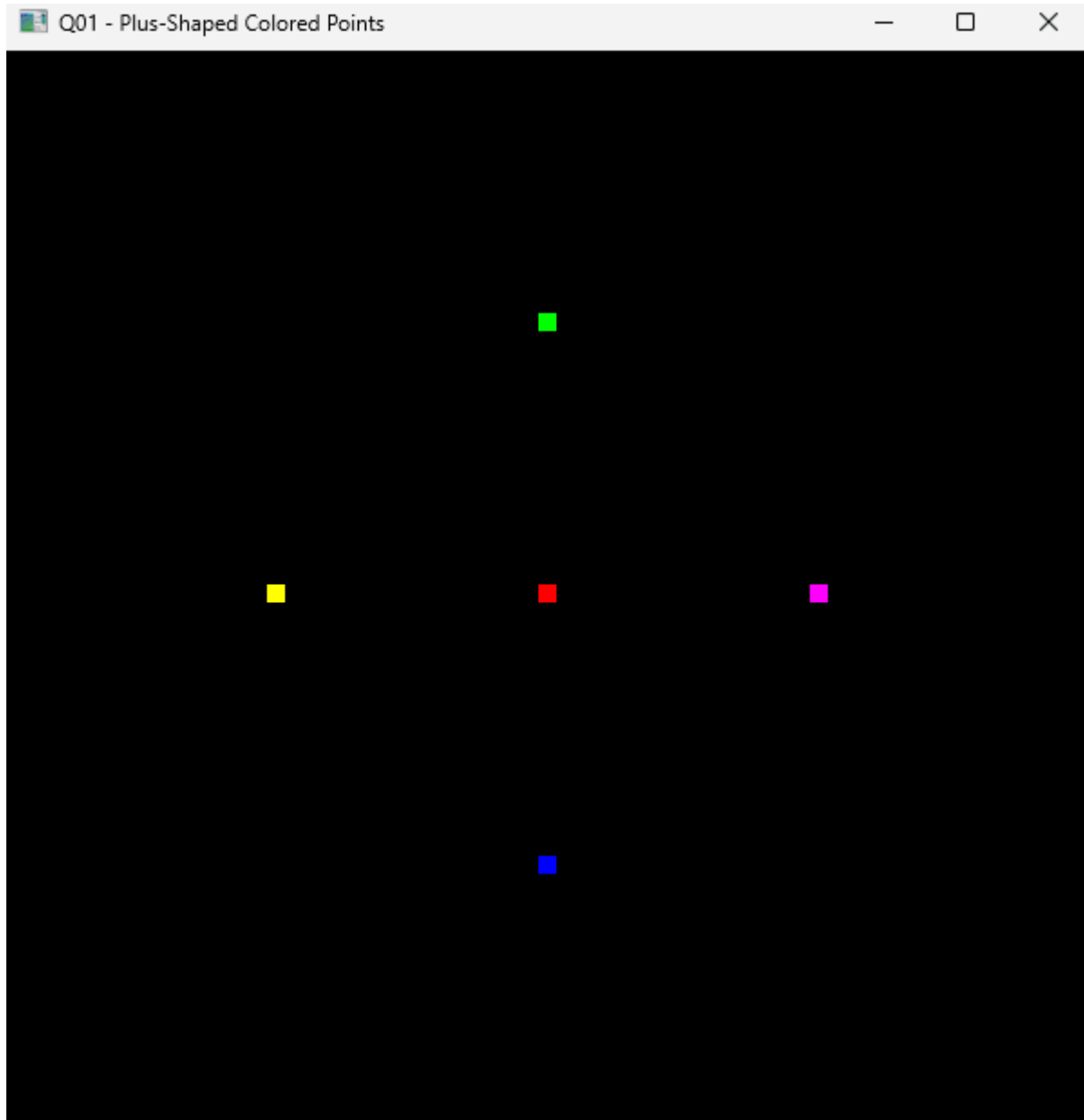
Toolchain: VS Code + MSYS2 UCRT64 GCC + FreeGLUT

Student Name:	Ebron, Nicole Anne V.
Section:	TN35
Professor:	Dr. Hadji J. Tejuco

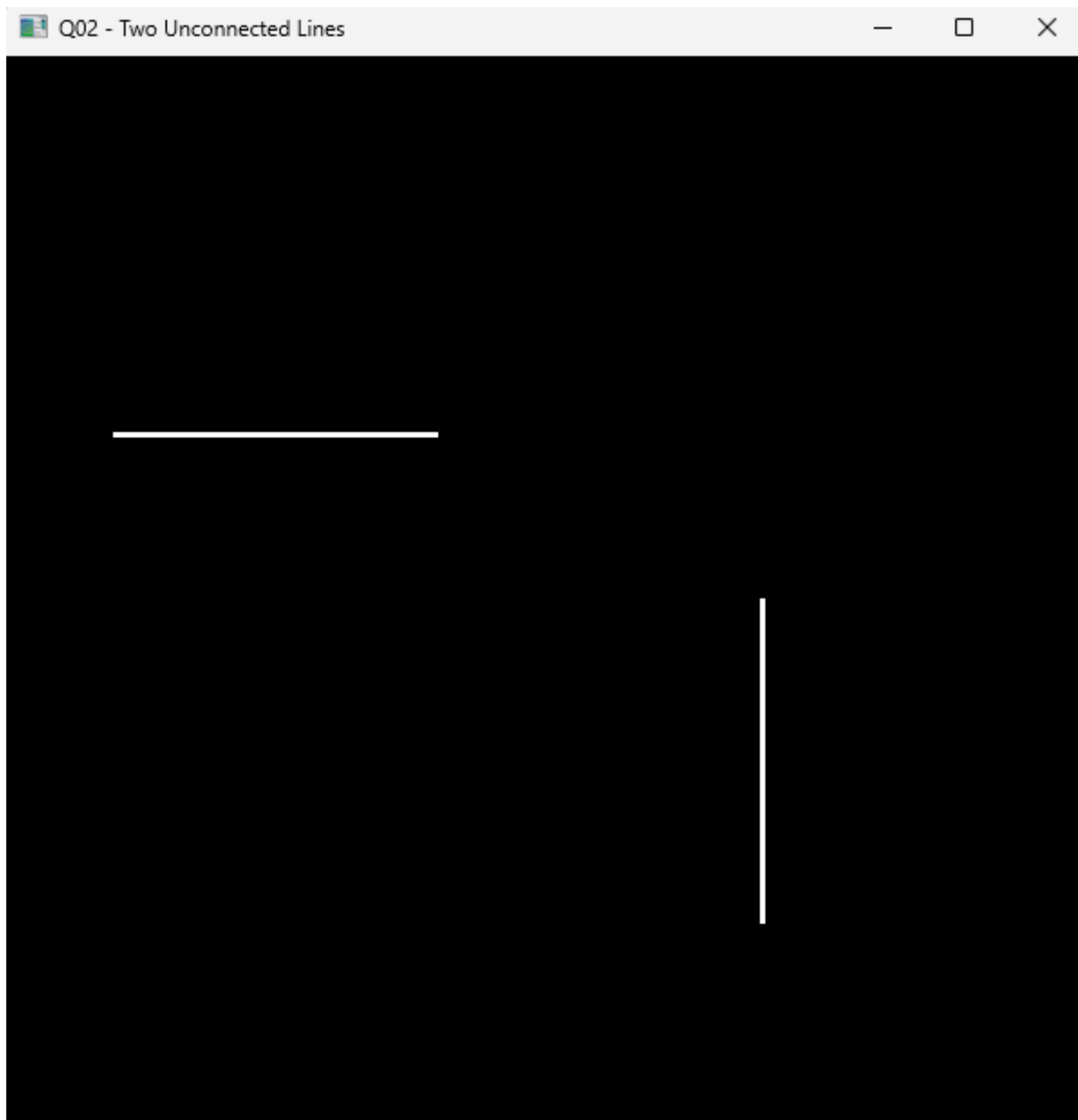
Part C — Outputs / Screen Shots (Ex01–Ex20)

Easy Exercises

Exercise Q01 — Plus-Shaped Colored Points [Easy]



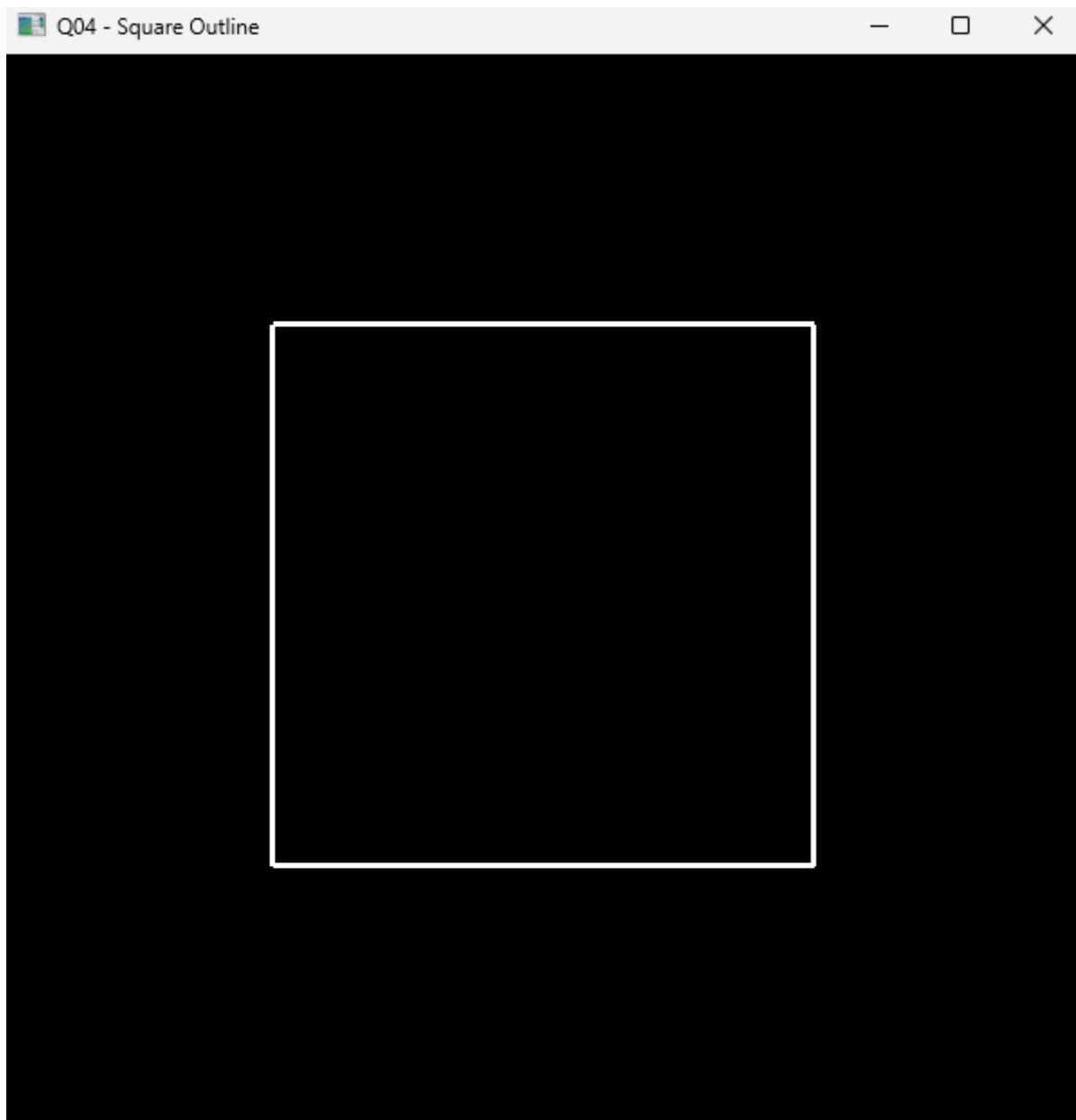
Exercise Q02 — Two Unconnected Lines [Easy]



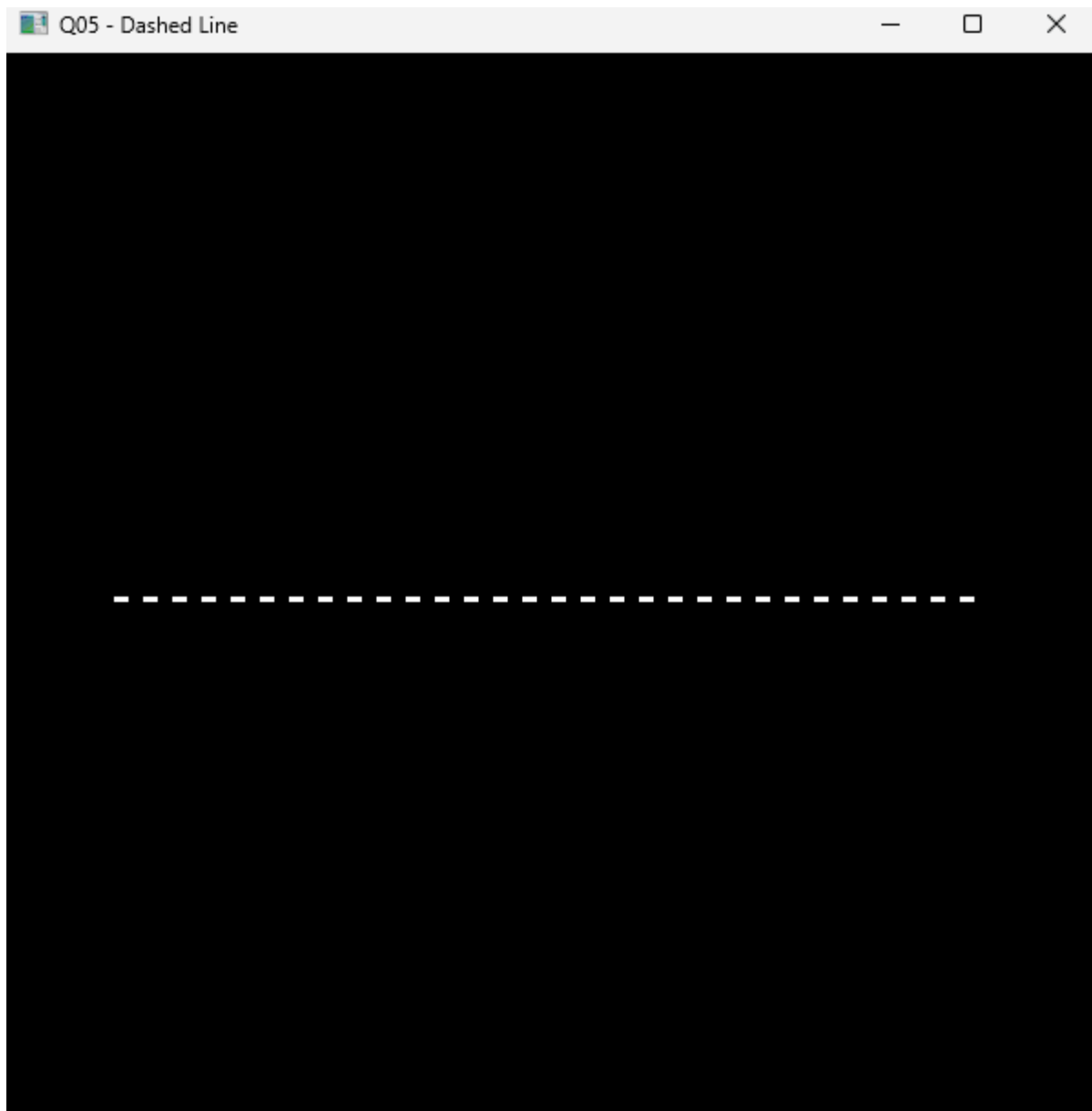
Exercise Q03 — Zig-Zag Line Strip [Easy]



Exercise Q04 — Square Outline (GL_LINE_LOOP) [Easy]



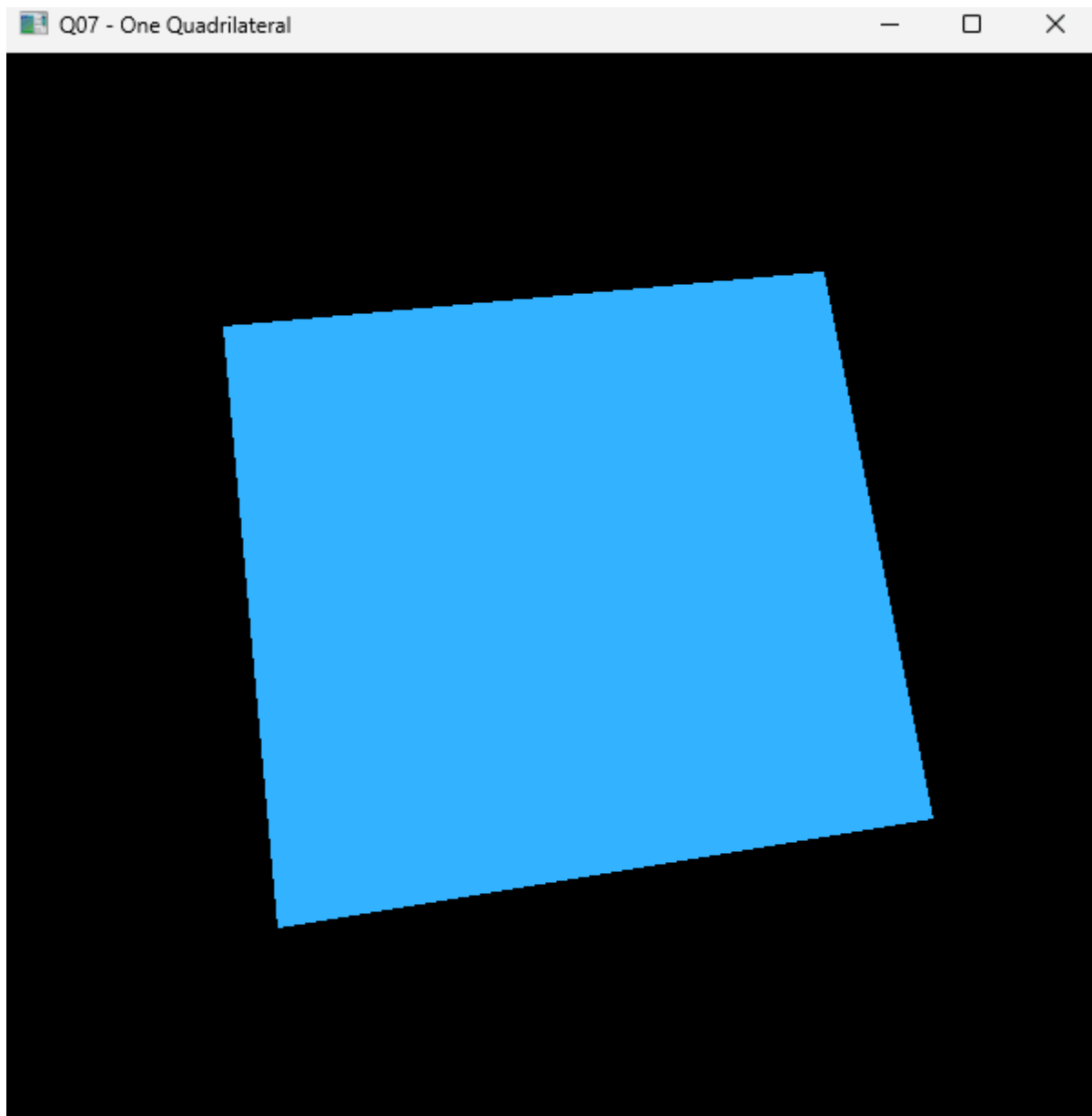
Exercise Q05 — Dashed Line [Easy]



Exercise Q06 — Triangle with glColor3ub [Easy]

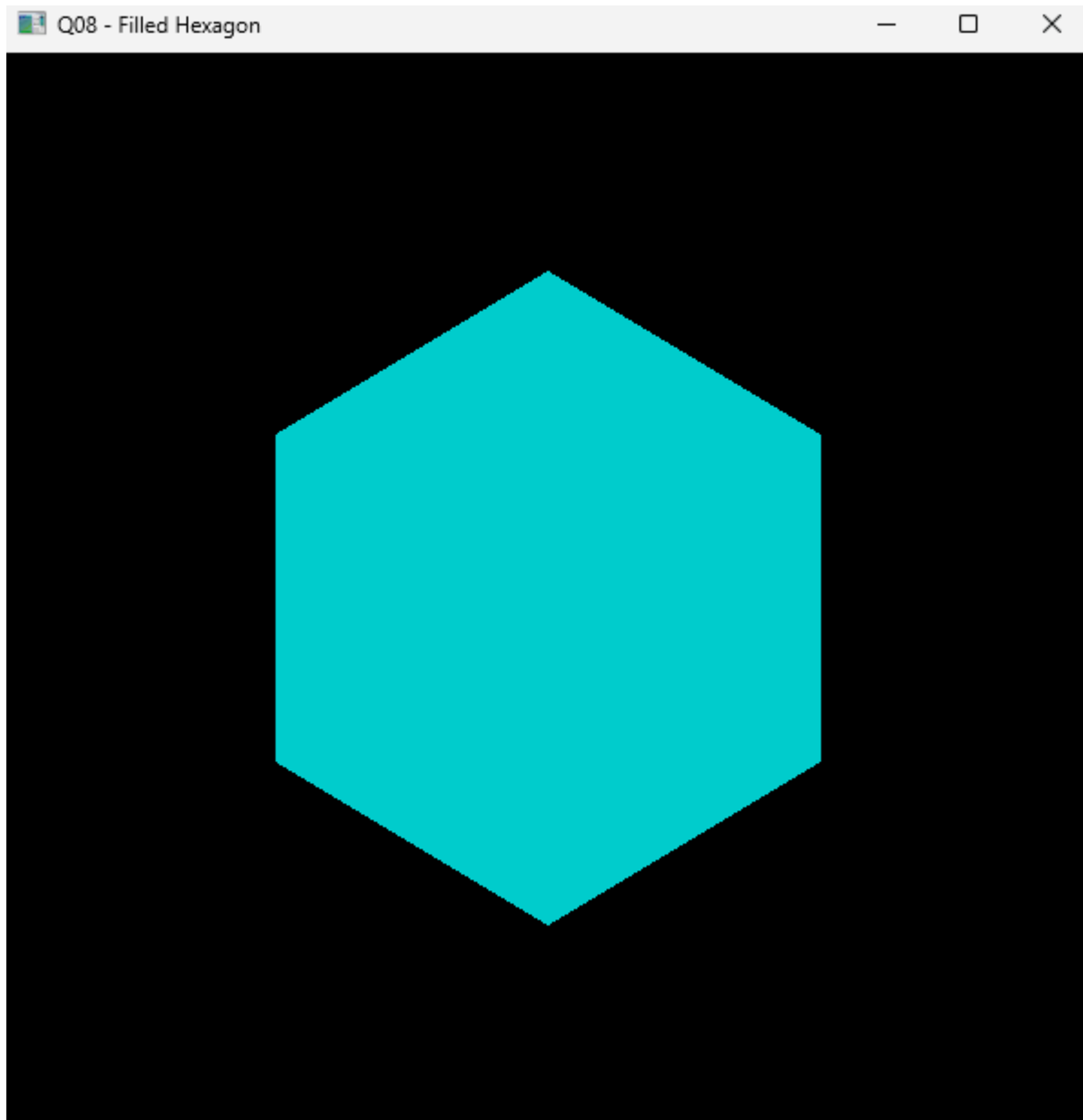


Exercise Q07 — One Quadrilateral (GL_QUADS) [Easy]

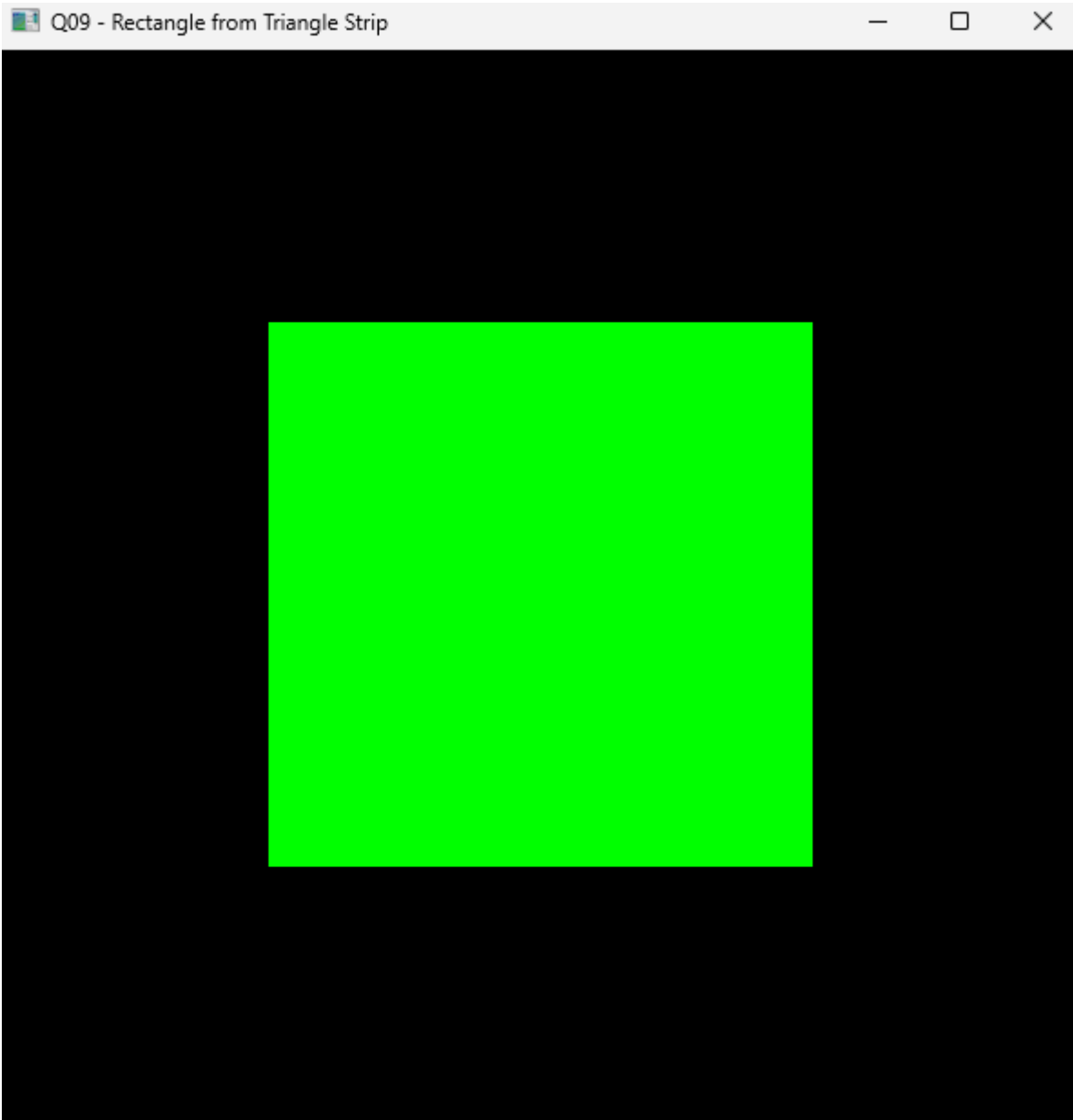


Medium Exercises

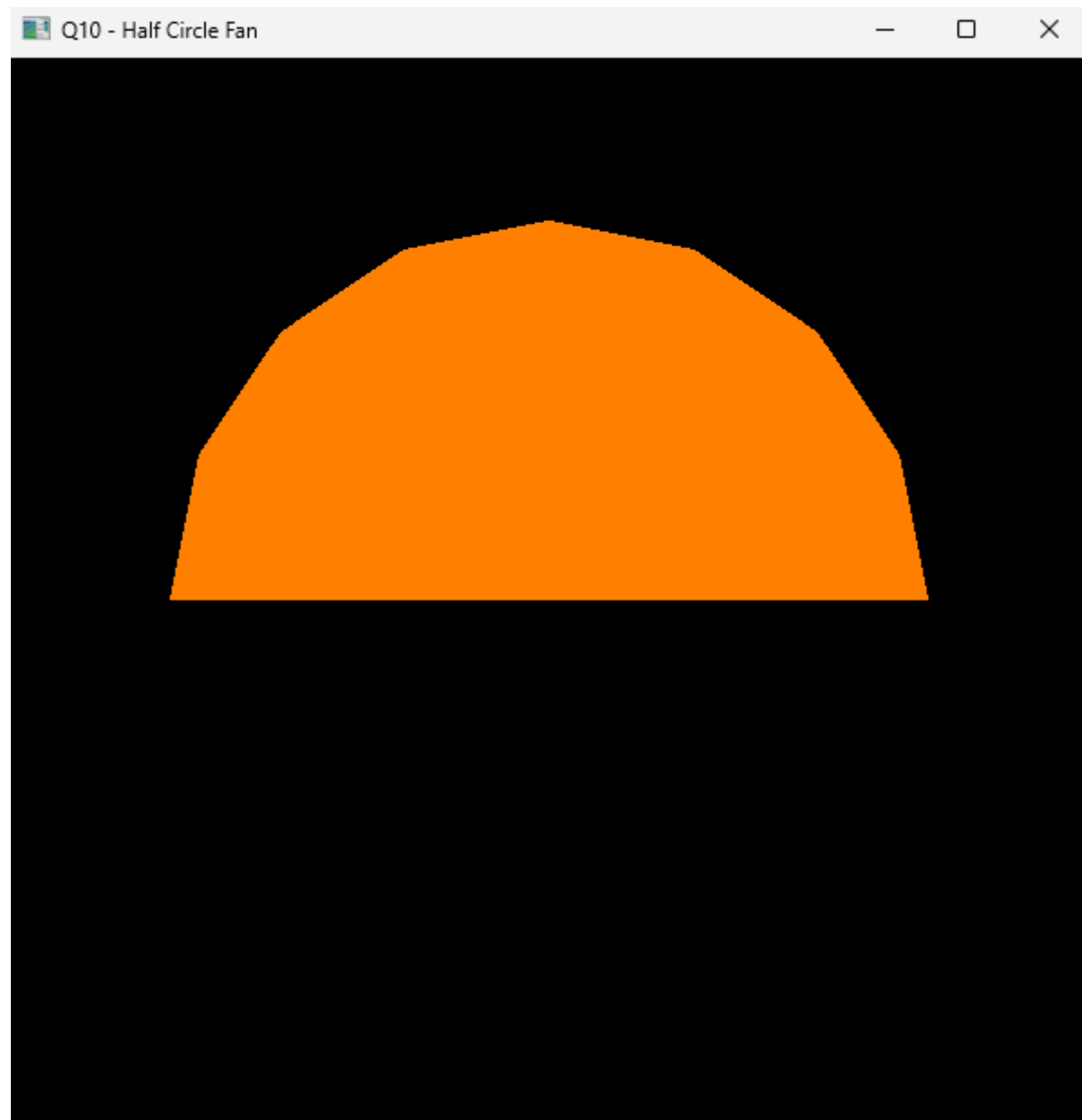
Exercise Q08 — Filled Hexagon (GL_POLYGON) [Medium]



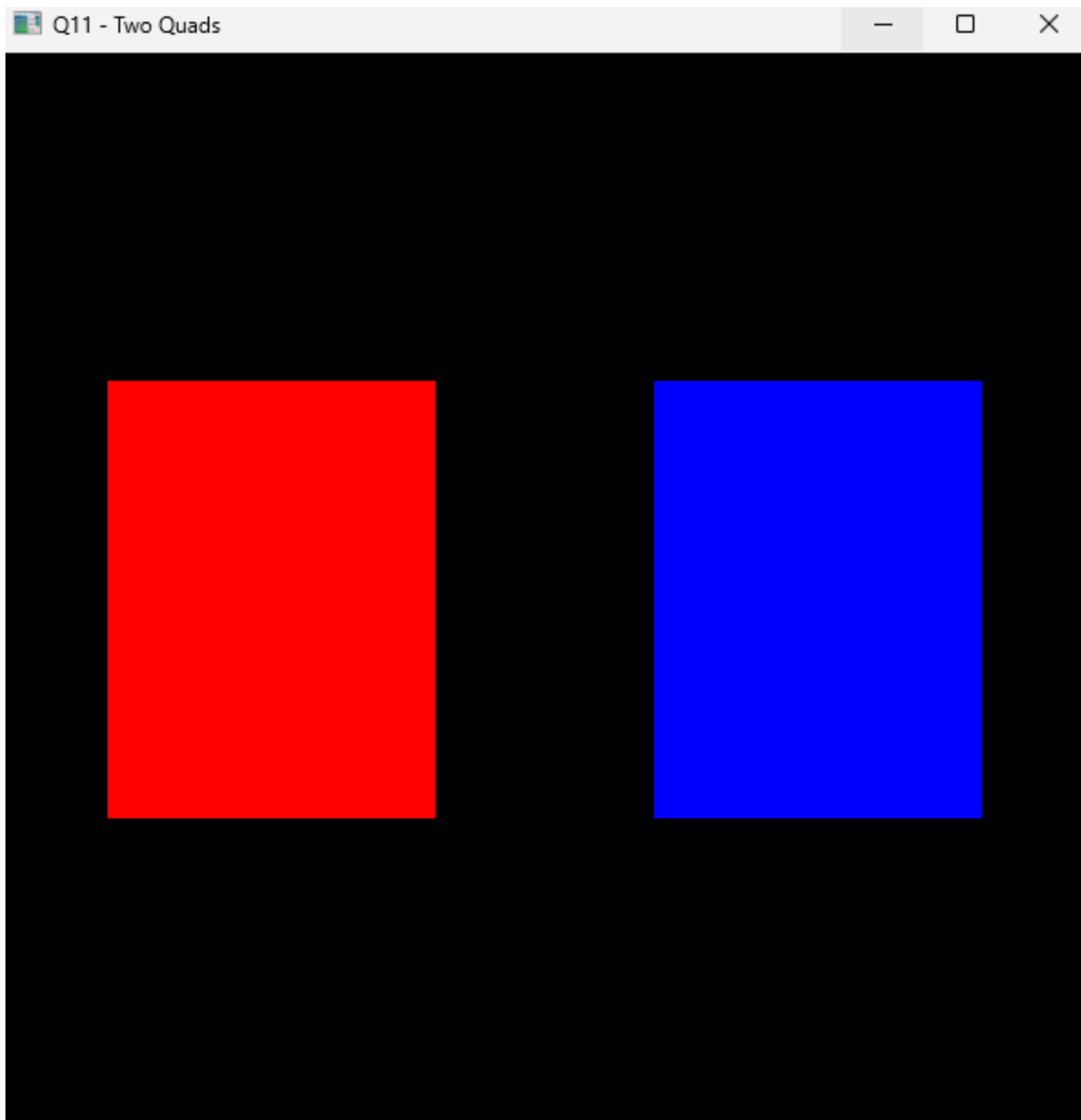
Exercise Q09 — Rectangle from Triangle Strip [Medium]



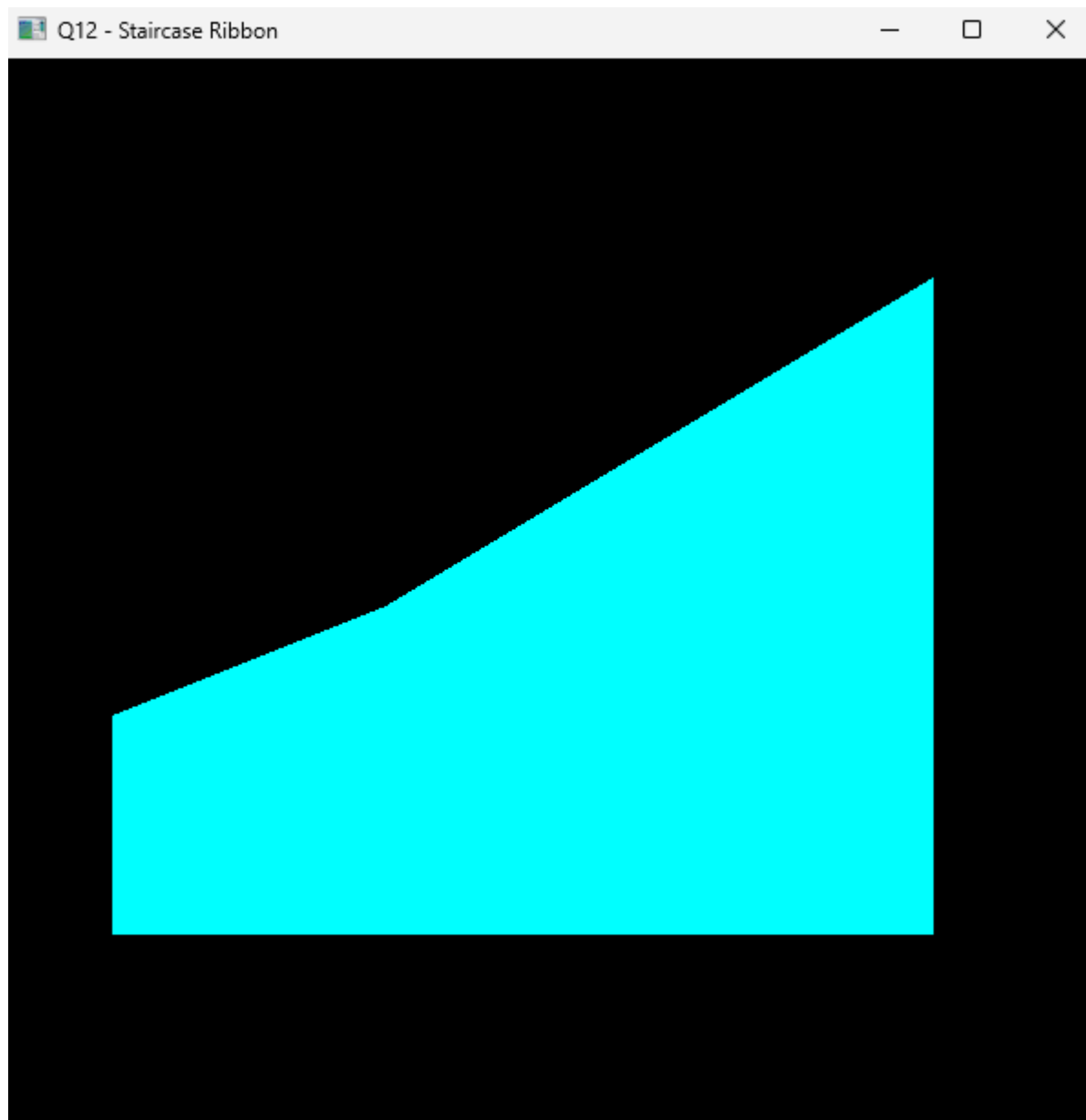
Exercise Q10 — Half-Circle Fan [Medium]



Exercise Q11 — Two Quads, One glBegin Block [Medium]



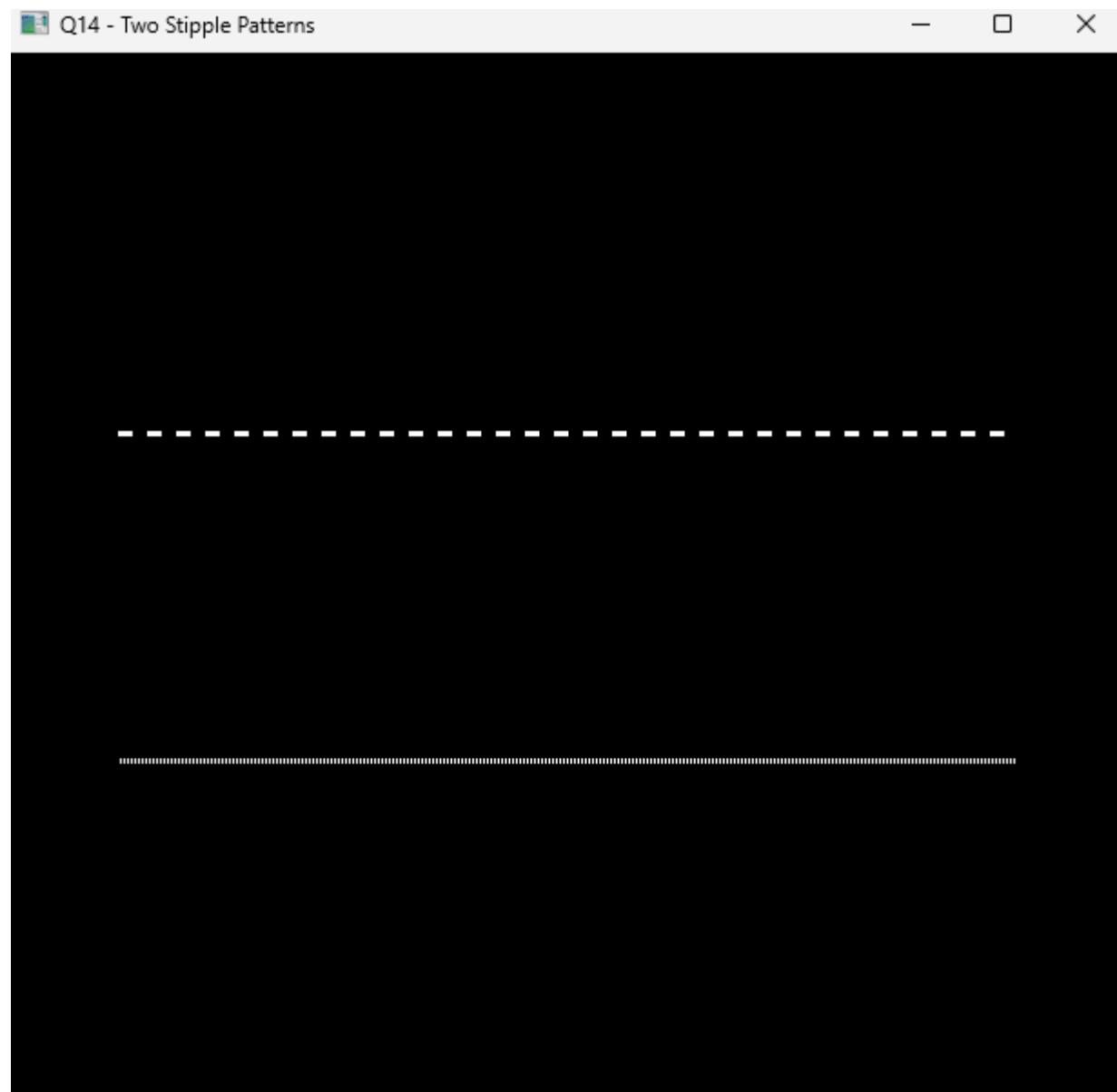
Exercise Q12 — Staircase Ribbon (GL_QUAD_STRIP) [Medium]



Exercise Q13 — Yellow-to-Purple Gradient Line [Medium]

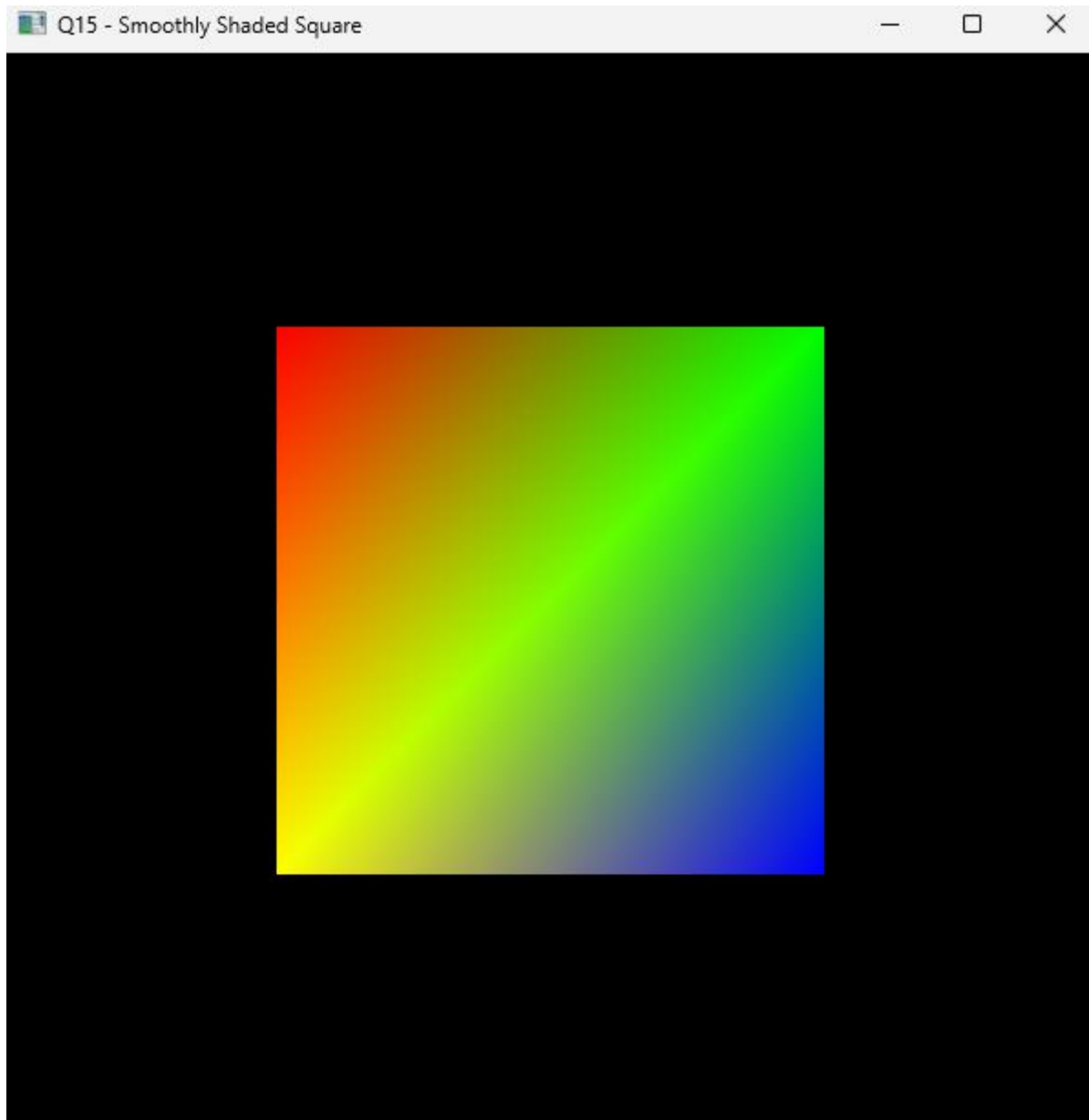


Exercise Q14 — Two Different Stipple Patterns [Medium]

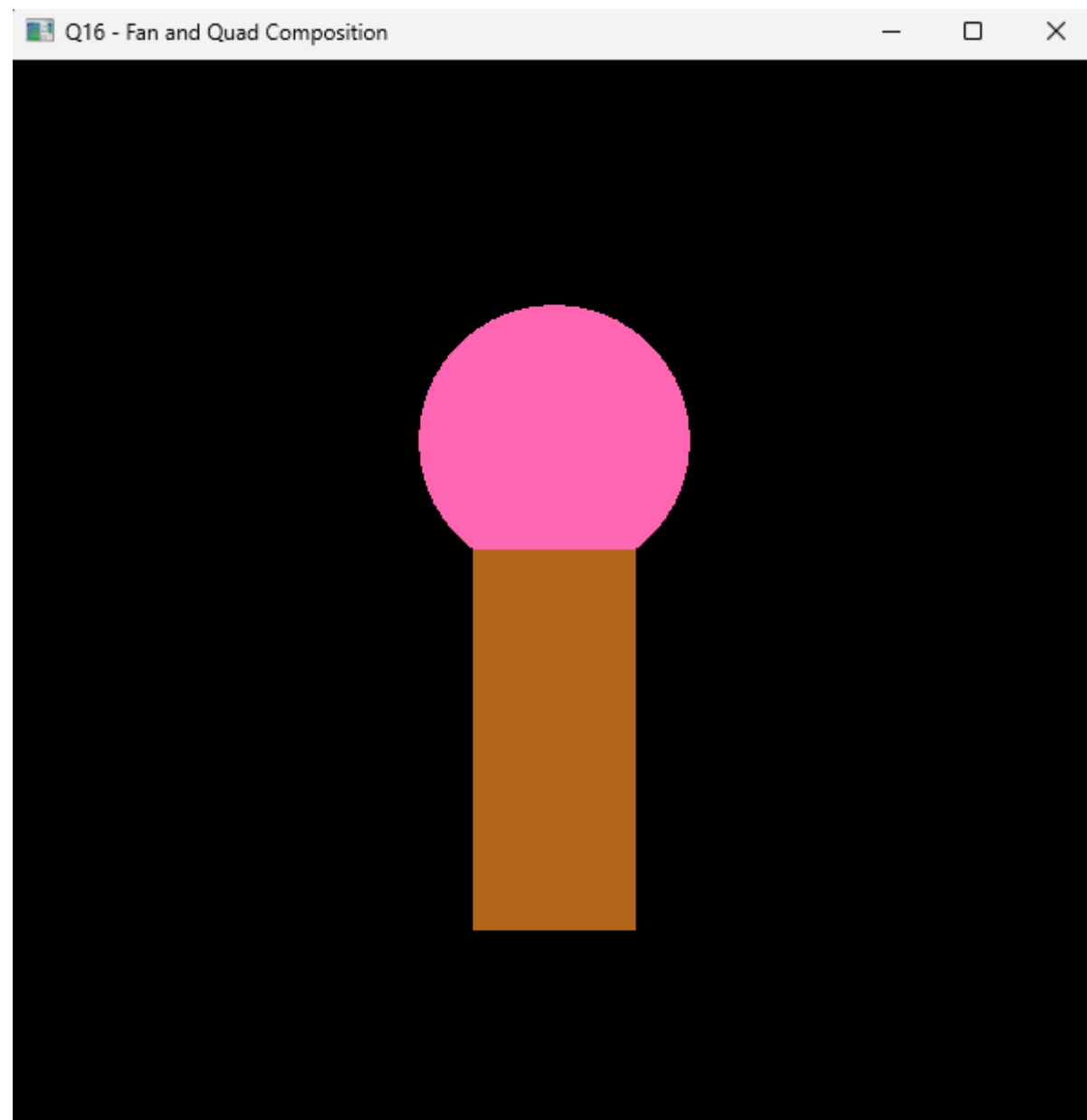


Hard Exercises

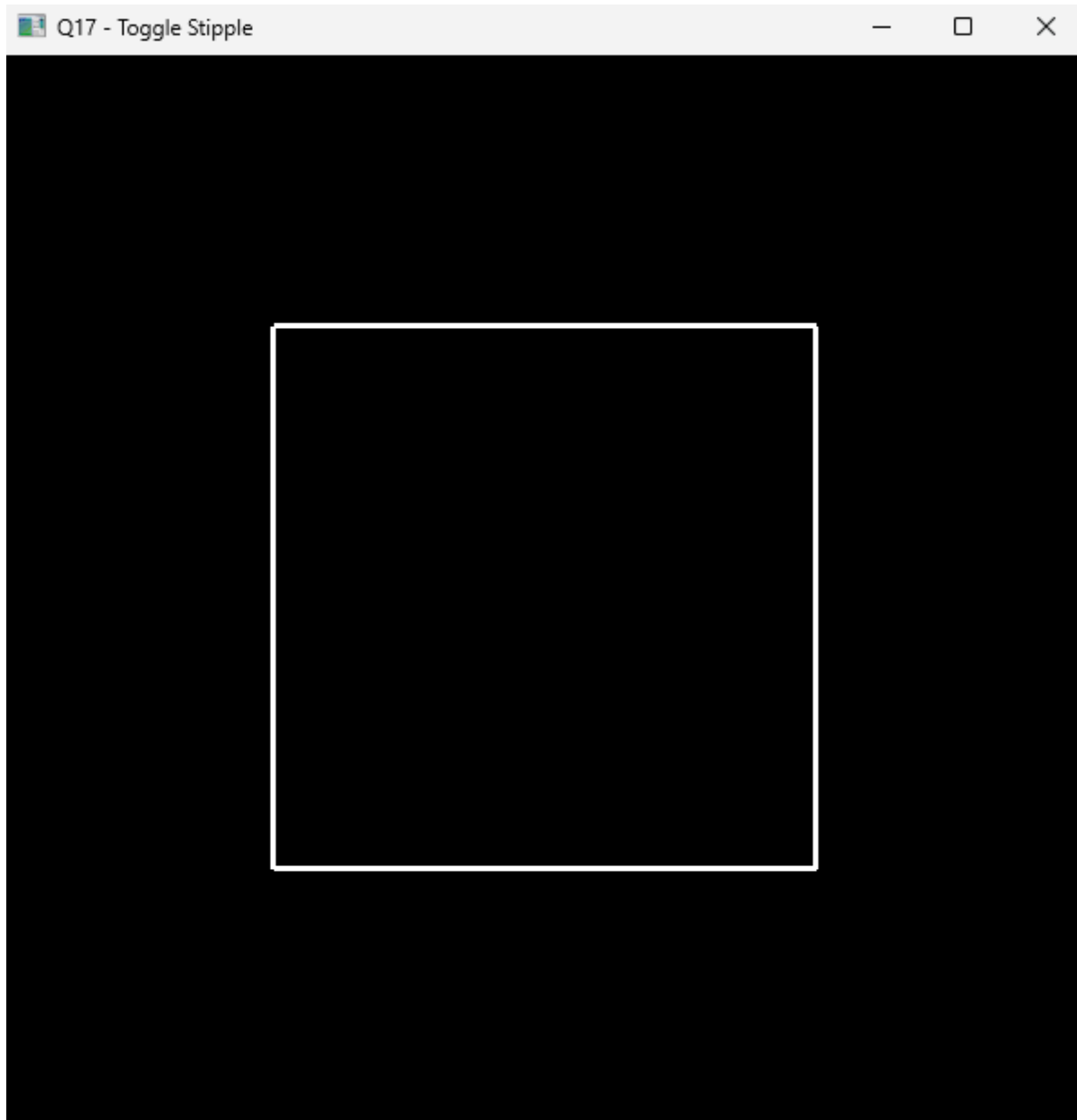
Exercise Q15 — Smoothly Shaded Square [Hard]



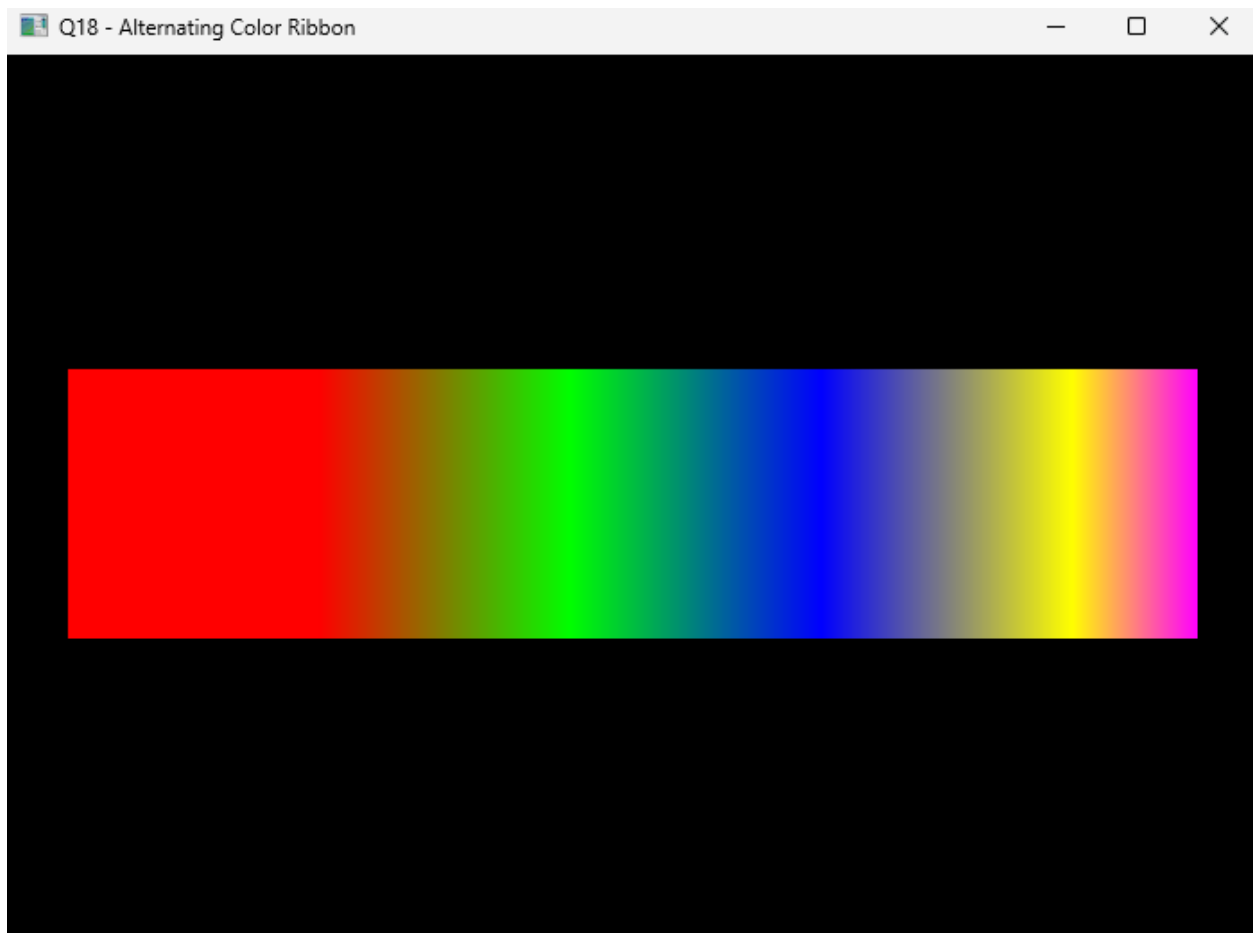
Exercise Q16 — Fan + Composition [Hard]



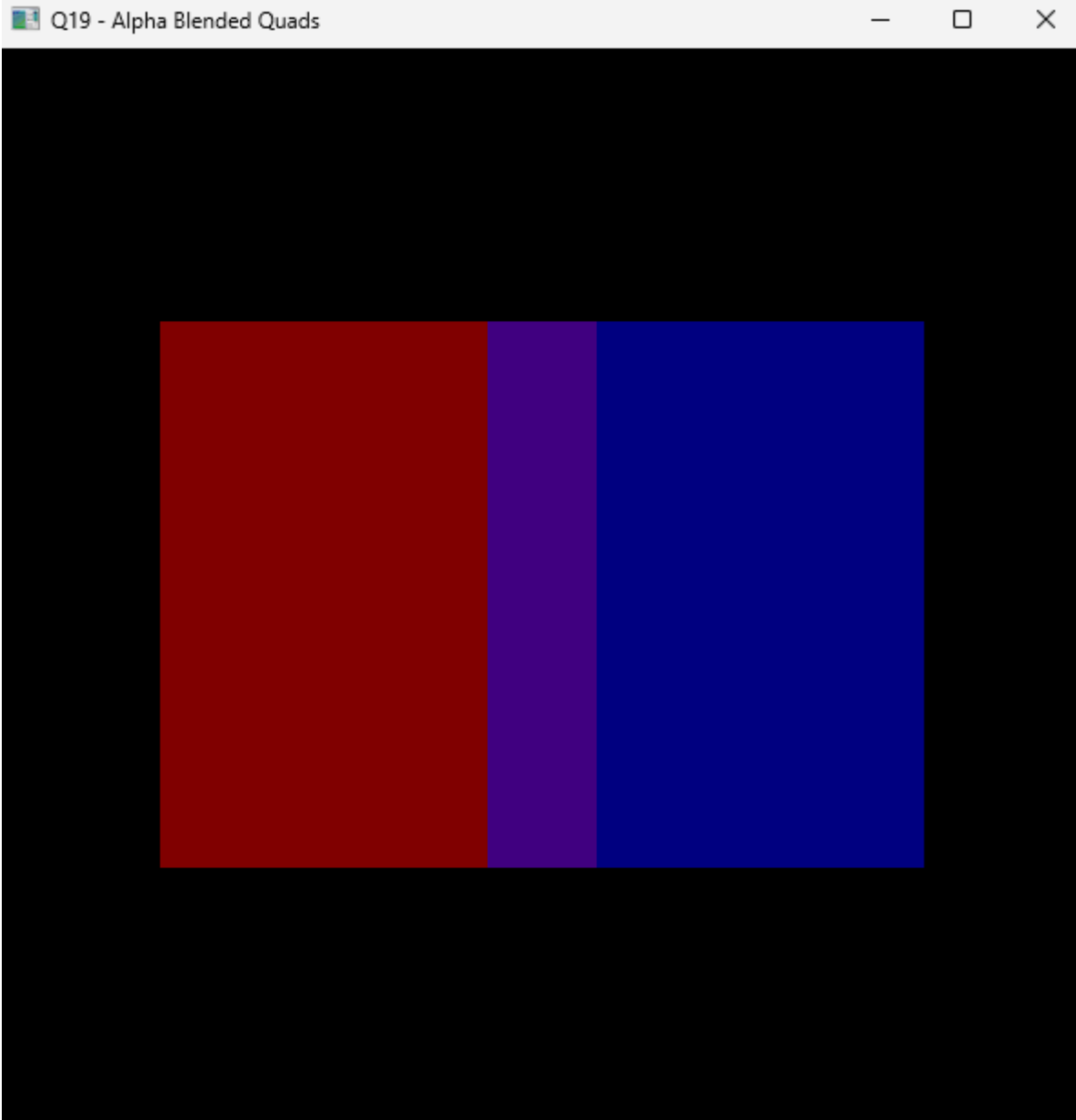
Exercise Q17 — Toggle Stipple with Keyboard [Hard]



Exercise Q18 — Alternating-Color Ribbon [Hard]



Exercise Q19 — Overlapping Alpha-Blended Quads [Hard]



Exercise Q20 — Procedural Rainbow Fan (Capstone) [Hard]

